वायु

Vayu

AI Decision Intelligence for Community Air Quality & Health — ask in plain language, get a fast, forecast-backed, explainable decision.

PARTICIPANT

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Solo submission

PROBLEM STATEMENT

AI for Better Living and Smarter Communities

Build an AI-powered Decision Intelligence Platform.

BRIEF ABOUT THE IDEA

Dashboards show numbers. People need decisions.

Every APAC megacity publishes air-quality data, yet a parent, a runner, or a city official still can't get a straight answer to the only question that matters: **"Given the air right now and where it's heading, what should I *do*?"**

REAL-WORLD PROBLEM & IMPACT

Air pollution is the top environmental health risk in APAC. Vayu converts open sensor data into **personal, population-aware health decisions** for citizens and a **situational-awareness + alert layer** for city stakeholders.

GOOGLE-CLOUD-FIRST APPROACH

A **Gemini**-powered assistant reasons over an analytics engine (AQI, forecast, anomalies); designed for **Vertex AI**, a **BigQuery** time-series lake, and one-command **Cloud Run** deploy. Grounded so it never invents numbers.

DATA → DECISION WORKFLOW

Ingest → compute **EPA AQI** → **forecast 24h** → **detect episodes** → synthesize a **recommendation** → narrate in natural language. A task that took an analyst minutes collapses to **one sentence in <1s.**

"Is it safe for my kid to play outside in Delhi this evening?" → "Delhi is 158 AQI (Unhealthy), driven by PM2.5, peaking at 218 (Very Unhealthy) around 8:30. For children, outdoor play isn't advisable now — a pollution spike was detected last night (PM2.5 207 vs 113 baseline)."

OPPORTUNITIES · USP & DIFFERENTIATION

Not another AQI dashboard — a decision engine.

HOW IT'S DIFFERENT

- Existing apps show a **number**; Vayu gives a **recommendation** tailored to who you are and what you plan to do.
- **Forward-looking**: a 24-hour forecast with a confidence band, not just the current reading.
- **Grounded & explainable** — the LLM only narrates engine-computed facts, with a "Why this answer?" panel. **No hallucinated numbers.**
- **Never goes dark**: provider-agnostic AI falls back to a local composer with zero cloud dependency.

CORE USP

The fastest path from **public air data** to a **trustworthy personal decision** — in one plain-language question.

🗣️ Plain-language assistant
🔮 24h forecast + band
🚨 Anomaly / episode detection
🧑‍⚕️ Population-aware advice
🛡️ Responsible / explainable AI
🌏 6 APAC cities

LIST OF FEATURES OFFERED BY THE SOLUTION

What Vayu does

Ask Vayu — NL assistant

Plain-English question → grounded answer + one clear recommendation, with best-time-window guidance.

Standard EPA AQI

PM2.5, PM10, NO_2, O_3 sub-indices with dominant-pollutant attribution — real domain logic.

24-hour forecast

Seasonal hour-of-day profile × recent trend, with a widening confidence band.

Pollution-episode detection

Robust rolling median + MAD z-score flags real spikes vs. baseline.

Population-aware decisions

Tailored guidance for children, elderly, asthma, pregnancy, and outdoor exercise.

Live cross-city alert feed

Unhealthy-air + episode alerts — the push layer for schools & city stakeholders.

PROCESS FLOW & USE-CASE

From a question to a decision, in one second

👤 **User asks** — "Safe for my kid outside in Delhi tonight?" → 🧭 **Intent parse** — city · profile (child) · activity → ⚙️ **Engine facts** — AQI · 24h forecast · episodes → 🧠 **Decision** — population-aware recommendation → 💬 **Gemini narrates** — grounded, explainable answer

CITIZEN USE-CASE

Parents, runners, commuters, elderly & asthma patients get an instant, personalized go / no-go call.

SCHOOL / WORKPLACE

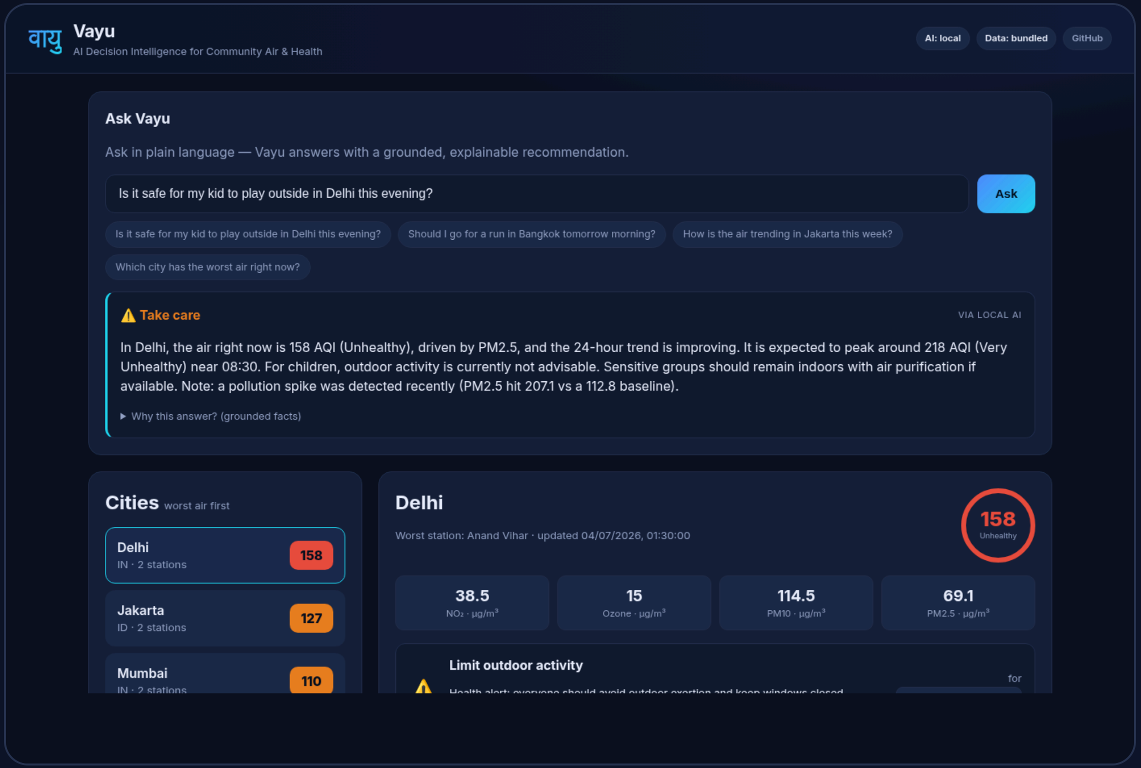
Auto-alerts drive decisions: move practice indoors, shift shifts, issue advisories.

CITY STAKEHOLDER

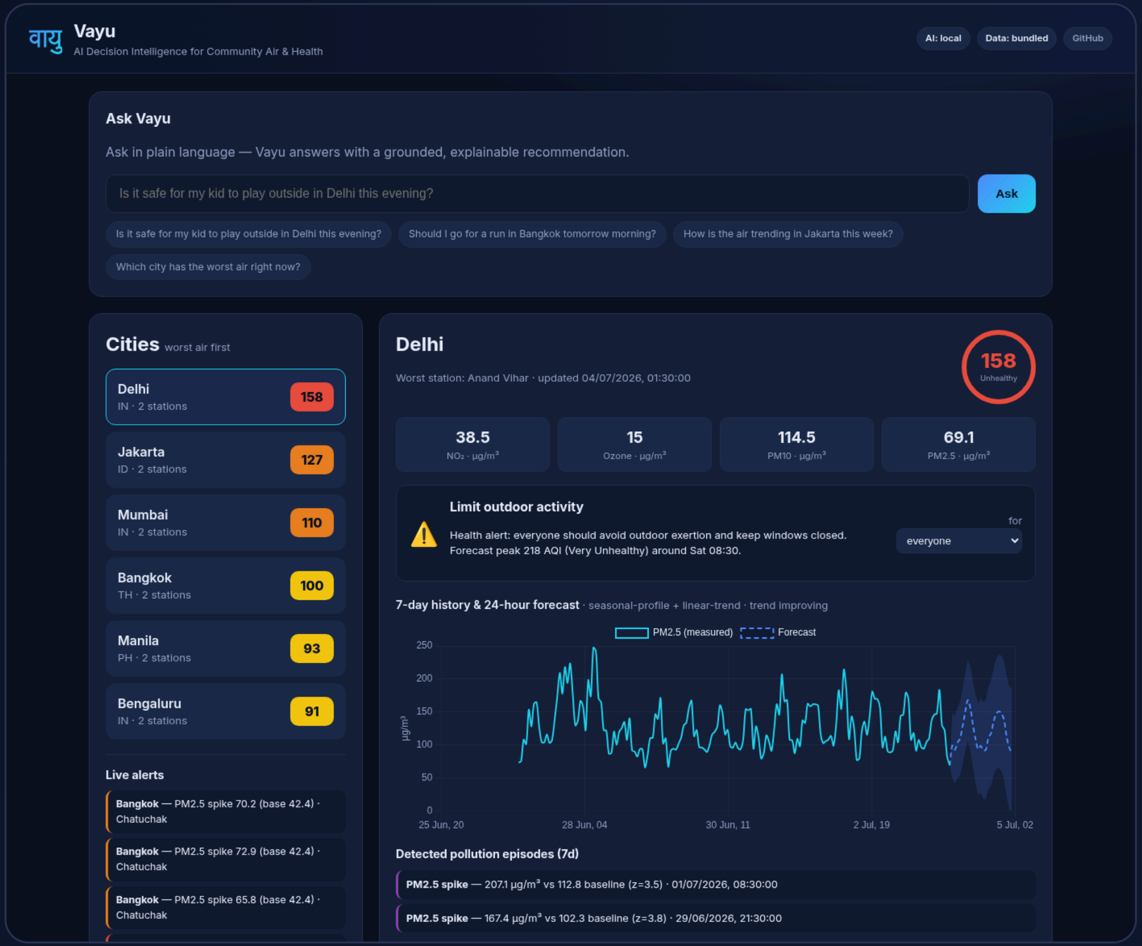
Cross-city episode feed + trends support faster interventions and public communication.

WIREFRAMES / MOCKUPS OF THE PROPOSED SOLUTION

One screen: ask, see, decide



Ask Vayu — natural-language answer with a verdict, AI-provenance badge, and an expandable "Why this answer?" panel.



Live dashboard — worst-air-first city list, AQI gauge, pollutant cards, population decision, 7-day history + 24h forecast, detected episodes, alert feed.

ARCHITECTURE OF THE PROPOSED SOLUTION

Grounded AI over a transparent analytics engine

DATA

Community sensors / OpenAQ → **Ingestion & enrichment** → **Time-series store — SQLite → BigQuery**

INTELLIGENCE ENGINE (EXPLAINABLE)

EPA AQI
sub-indices + dominant

+

Forecast
seasonal × trend + band

+

Anomaly
MAD z-score episodes

→

Decision engine
population-aware rules

AI & DELIVERY

Provider-agnostic LLM
Gemini / Vertex AI · OpenAI · local fallback

→

Flask API
REST · grounded prompts

→

Dashboard
Chart.js SPA

→

Cloud Run / VM
containerized, HTTPS

TECHNOLOGIES & GOOGLE CLOUD SERVICES

Why this stack — built to scale & deploy for real

🔷 Gemini / Vertex AI

Natural-language reasoning + narration. Grounded on engine facts for trustworthy, explainable answers. `google-genai` wired in; flip on with an API key.

🔶 BigQuery

The station × pollutant × time schema maps directly to a BigQuery lake for city-scale, multi-year, low-latency analytics.

🚀 Cloud Run

Containerized (Dockerfile included) for one-command, autoscaling, scale-to-zero deployment — cost-efficient for public services.

🐍 Python · Flask · NumPy

Lightweight, ARM-friendly engine (EPA AQI, forecasting, anomaly detection). Runs on a 2-vCPU node or scales horizontally.

📈 Chart.js (vendored)

Zero external runtime JS dependency — the app is fully self-contained and works offline.

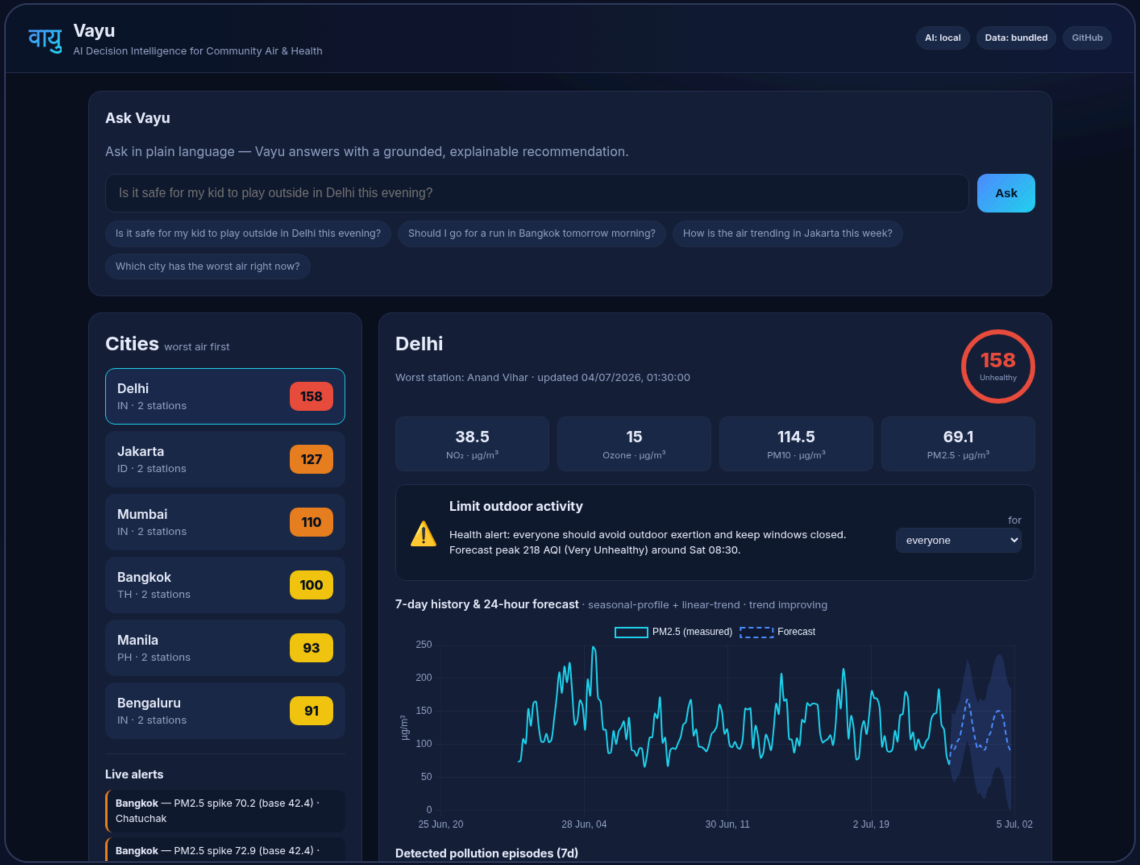
🌐 OpenAQ + Cloudflare

Optional live real-world data; deployed today over HTTPS. Provider-agnostic AI keeps the demo resilient.

Scalability & real-world deployment: stateless API + managed data store → horizontal scale on Cloud Run; the engine is a civic re-point of a **production real-time analytics platform** we run for financial markets — the time-series shape is identical, so forecasting & anomaly detection transferred directly.

SNAPSHOTS OF THE PROTOTYPE

Live, deployed, and working today



LIVE PROTOTYPE

🔗 admit-repository-qualified-administrative.trycloudflare.com

SOURCE CODE (PUBLIC)

🔗 github.com/Lozsku/vayu-air-intelligence

6
APAC cities

34k+
measurements

<1s
to a decision